OptArrow Process Documentation

1.1 Business or mission analysis process

Process ID	**TEC. 1**
Process Reference	ISO/IEC/IEEE 12207:2017, Clause 6.4.1 (Business or Mission Analysis)
Process name	Business or mission analysis process
Process purpose	The purpose of this process is to define the problem space for cross-language optimization and characterize a solution that leverages Apache Arrow for high-performance integration between Python and Julia ecosystems.
	Process Outcomes and Source
Base Practices (PO)	**PO. A: The problem or opportunity is defined** **Description:** This describes optimization problem between two-languages (Python and Julia). **Problem Statement:** Researchers often prototype in Python but require Julia (and JuMP) for high-performance modeling and solver access. Existing integration methods (like PyJulia) suffer from complex environment management, serialization bottlenecks, and memory overhead when transferring large sparse matrices. **Source:** README.md (Line 6- 7) and Architecture.md (Line 10- 25**)** **Opportunity:** Utilize Apache Arrow as a universal, columnar memory format to provide a "zero-copy" or low-overhead IPC mechanism for optimization data (A-matrices, cost vectors, bounds). **PO. B: The solution space is characterized** **Description:** This defines the boundaries and constraints of the OptArrow engine. • **Solution Scope:** A distributed computation service comprising an HTTP Gateway, a Python Engine (via PyArrow Flight), and a Julia Engine (via TCP/Arrow Stream).

	• **Feasibility/Constraints:** Must support sparse matrix formats (COO/CSC); must remain language-agnostic; target environment: Docker. Source: Architecture.md and compose.yaml **PO. C:** Preliminary life cycle concepts are defined **Description:** This defines the preliminary life cycle concept of OptArrow. It describes how the system will be developed, deployed, and operated. **Operational Concepts:** • Stateless Mode: Current operational state for independent solves. • Iterative Mode: Planned evolution for sequential dependent optimization. Source: Architecture.md **PO. D:** Preliminary requirements are defined **Description:** High-level capabilities required to meet the mission. **Functional:** Support for LP and QP model types; integration with GLPK, Gurobi, HiGHS, Mosek. **Source Reference:** README.md (L12-L19) and Architecture.md (L137-L162)

Base Practice (BP)	ISO 33061 Process	Summary

	Assessment Model	
BP1: Has the problem or opportunity been defined?	Fully Achieved (F)	Defined in README.md as the "two-language problem."
BP 2: Has the solution space been characterized?	Fully Achieved (F)	Documented in Architecture.md; scope limited to Arrow-based IPC.
BP 3: Have preliminary life cycle concepts been defined?	Largely Achieved (L)	Stateless concepts defined; Iterative concepts are planned but not fully implemented.
BP4: Have preliminary requirements been defined?	Fully Achieved (F)	High-level requirements (LP/QP, specific solvers) are documented.
BP 5: Have stakeholders and their needs been identified?	Fully Achieved (F)	Stakeholder table in this document maps roles to specific performance/usability needs.

1.2 System/ Software requirement definition process